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DATA MINING

FINAL REPORT

Topic: Air pollution early warning

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Abstract

Average PM2.5 concentrations in Beijing have fallen, yet severe episodes persist, and most forecasting studies predict concentrations rather than the onset of an episode. This proof-of-concept study asks whether the onset of a PM2.5 episode (at least six consecutive hours above 75 $\mu g/m^3$) can be anticipated from information available up to the current hour, and whether the model transfers to unseen stations. We analysed 420,768 hourly records from 12 Beijing stations (March 2013–February 2017). PM2.5 peaked in winter, but episodes occurred in all months; co-pollutants were strongly associated with PM2.5, weather variables weakly, and two weather regimes did not separate onsets. LightGBM classifiers were trained on seven input scenarios using a chronological split and a cap of 12 false alarms per station per 30 days. The pre-specified hypothesis that co-pollutants and weather without PM2.5 would outperform a PM2.5-only model was not supported (episode recall, warning $\geq$ 1 h ahead, at H = 6 h: 0.70 vs 0.90). A model combining PM2.5 and weather at H = 12 h reached a usable recall (warning $\geq$ 3 h ahead) of 0.81 with about 11 false alarms per station per 30 days, only 0.047 above PM2.5-only in an exploratory comparison. In leave-one-station-out validation, usable recall fell by just 0.013 and all 12 stations passed the transfer gate. Findings come from one city and one test year and require further validation before operational use.

Keywords: PM2.5 episodes; Early warning; Episode onset prediction; LightGBM; Leave-one-station-out validation; Spatial generalization; Beijing.

Code Repository: YOUR_GITHUB_LINK

1 Introduction

1.1 The problem and its urgency

Air pollution is a serious environmental and public health issue in many urban areas. Among various air pollutants, PM2.5 is an important indicator because these fine particles can penetrate deep into the respiratory system and adversely affect human health [15]. In Beijing, although air quality has improved due to emission control policies, PM2.5 pollution continues to fluctuate across time, seasons, locations, and meteorological conditions. Wang et al. [14] reported that the average PM2.5 concentration decreased from 99.4 to 69.4 $\mu g/m^3$ between 2013 and 2017. However, a reduction in average concentration does not necessarily mean that severe pollution episodes have been eliminated. Therefore, beyond assessing overall pollution levels, it is essential to identify when a pollution episode begins and whether its occurrence can be anticipated before PM2.5 exceeds the threshold.

1.2 Research gap

Existing studies have paid limited attention to predicting the onset of sustained pollution episodes before the threshold is reached. In addition, conventional forecasting metrics may not adequately reflect practical early-warning performance, while spatial generalizability across unseen monitoring stations remains insufficiently addressed [6, 9].

1.3 Research objectives

This study aims to achieve two main objectives:

- Analyze temporal, meteorological, and pollutant-related conditions associated with high PM2.5 concentrations.
- Develop a model to anticipate the onset of pollution episodes using information from previous hours and evaluate its transferability across monitoring stations.

Based on these objectives, one hypothesis was specified in advance: a model using co-pollutants and weather without PM2.5 would outperform a PM2.5-only model in both PR-AUC lift and recall under the same false-alarm budget. This study is conducted as a proof of concept rather than an attempt to develop a warning system that is ready for real-world deployment.

1.4 Research questions

Based on the research objectives, five research questions are developed, following a sequence from observing pollution patterns and examining related factors to predicting episode onset and evaluating the model's generalizability across monitoring stations.
Master research question:
Can historical data on time, meteorological conditions, and air pollutants identify the onset of high-PM2.5 pollution episodes, and can the model generalize across monitoring stations?
Specifically, this study addresses five research questions:

- ***RQ1*** – ***Observe:*** When and at which monitoring stations do PM2.5 concentrations reach their highest levels? When do pollution episodes typically occur?
- ***RQ2*** – ***Explain:*** Which meteorological factors and co-pollutants have the strongest associations with PM2.5?
- ***RQ3*** – ***Discover:*** Do distinct weather-pollution regimes exist, and how are pollution episodes distributed across these regimes and seasons?
- ***RQ4*** – ***Identify:*** Can episode onset be identified within the next H hours using historical data? How does model performance change across different lead times?
- ***RQ5*** – ***Generalize:*** Does the model perform well at stations not included in the training data, or are station-specific models necessary?

1.5 Contribution

This study makes three main contributions:

- **Methodological contribution:** Shifts the focus from forecasting PM2.5 concentrations to identifying episode onset within the next 1, 3, 6, 12, and 24 hours, while restricting predictions to periods when PM2.5 remains below the threshold.
- **Decision-support contribution:** Combines PR-AUC, episode recall, usable recall, false alarms, and median lead time to evaluate the practical usefulness of early warnings. It also examines the cost trade-off using an assumed 10:1 ratio between missed episodes and false alarms.
- **Spatial generalization contribution:** Applies Leave-One-Station-Out (LOSO) validation to examine transferability across the 12 monitoring stations, thereby determining whether a general model is suitable or station-specific models are necessary.

2 Related work

2.1 PM2.5 pollution trends in Beijing

Wang et al. [14] reported a significant decrease in average PM2.5 concentrations in Beijing between 2013 and 2017. Shi et al. [12] also confirmed that China's air pollution control policies during 2013–2020 contributed to emission reductions. However, severe pollution episodes still need to be examined separately rather than relying solely on average concentration levels.

2.2 The influence of meteorological conditions

Factors such as temperature, humidity, pressure, and wind influence the dispersion and accumulation of PM2.5 [5]. Zhang et al. [16] demonstrated that the effects of wind do not follow a simple linear relationship: maximum wind speed played a significant role, whereas average wind speed did not show a clear effect in the model examined. These findings motivate examining combinations of meteorological conditions jointly, here through clustering techniques.

2.3 PM2.5 forecasting studies and event-based early warning

Previous PM2.5 forecasting studies have mainly focused on predicting future concentration levels using machine learning and deep learning approaches, including CNN, LSTM, and BiLSTM architectures [1, 3, 13]. Beyond continuous concentration prediction, some studies have addressed pollution forecasting as a classification problem. For example, Logothetis et al. [4] used LSTM networks to forecast PM2.5 threshold exceedances, while Miskell et al. [8] applied gradient-boosted models to predict short-term PM2.5 peak events. Saiohai et al. [10] further investigated early warning of excessive PM2.5 episodes using vertical meteorological factors and machine learning.
Despite these advances, predicting concentration levels, threshold exceedances, or short-term peaks does not necessarily identify the onset of a sustained pollution episode before the threshold is reached. In particular, forecasting a temporary exceedance differs from anticipating the beginning of a prolonged pollution event. Based on the studies reviewed, an important research gap remains in predicting the onset of sustained multi-hour PM2.5 episodes in Beijing while observed concentrations are still below the warning threshold. This study addresses this gap by defining an episode threshold of 75 $\mu g/m^3$, corresponding to the 24-hour Grade II limit under China's Ambient Air Quality Standards [7].

2.4 Research gap

Based on previous studies, three main research gaps are identified:

1. **Prediction gap:** Previous studies primarily focus on forecasting PM2.5 concentrations or threshold exceedances, while less attention has been paid to identifying episode onset when PM2.5 remains below the threshold and no episode has started.

2. **Evaluation gap:** Model performance needs to be evaluated in terms of practical early-warning capability through episode-level recall, false alarms, and lead time, in addition to conventional forecasting metrics.

3. **Spatial gap:** Using data from multiple monitoring stations does not necessarily demonstrate that a model can perform well at stations that were not included in the training data. Spatial dependence can make validation that mixes locations overly optimistic, which is why validation that withholds whole locations is recommended [6, 9].

This study aims to address these three gaps through episode-onset classification, cost-based performance evaluation, and spatial generalization testing across monitoring stations.

3 Methodology

3.1 Data and preprocessing

Hourly records from 12 Beijing monitoring stations were used (1 March 2013 to 28 February 2017; 35,064 hours per station, 420,768 station-hours in total). The variables were PM2.5 (target), five co-pollutants (PM10, SO2, NO2, CO, O3), five weather variables (temperature, pressure, dew point, rainfall, wind speed), and wind direction, used only in RQ4–RQ5 (Table 1). A complete hourly grid was built for each station. No rows were removed and no values were imputed; missing values were kept as NaN. Seasons followed calendar months: winter (December–February), spring (March–May), summer (June–August), and autumn (September–November). The workflow is shown in Fig. 1.

Table 1: Scenarios and the reason each is included

Scenario	**Inputs at time t**	**Features**	**Why It Is Included**
Climatology	None (station × month × hour frequency)	0	Floor: what the calendar alone predicts. Any model must beat it.
PM2.5-only	PM2.5(t), PM2.5($t-1$)	2	Hardest cheap baseline: operators already see current PM2.5. RQ4 is only useful if it adds to this.
A: Pollutants	Lagged PM10, SO2, NO2, CO, O3 (no PM2.5)	25	Do co-pollutants carry early information (RQ2)? Uses other sensors.
W: Weather	Wind, temperature, pressure, calendar	27	Does meteorology alone anticipate onset? Inputs are forecastable and sensor-free.
B: A + W	A and W (still no PM2.5)	52	Most information without PM2.5. Pre-specified primary test: B vs PM2.5-only.
PW: W + PM2.5	W and PM2.5(t), PM2.5($t-1$)	29	Practical candidate: one PM2.5 sensor plus a weather feed. Exploratory.
C: B + PM2.5	B and PM2.5 lags/changes	58	Information ceiling and leakage check (C must not be worse than B).

Note: All features use information up to hour t only. No station identifier is used, so every model can be applied to a new station.

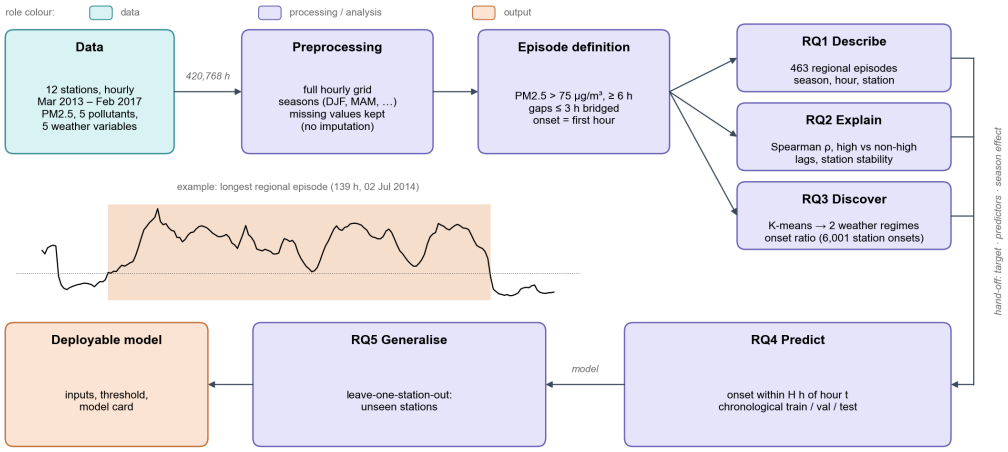


Figure 1: Research Workflow

3.2 Definition of PM2.5 episodes and onsets

A PM2.5 episode was defined as a run of consecutive hours with PM2.5 $> 75\ \mu g/m^3$ lasting at least 6 hours; the onset was its first hour. Two gap-handling rules were used. For descriptive analyses (RQ1–RQ3), runs separated only by missing data (at most 3 hours) were merged. RQ1 used the regional series (mean across available stations), and RQ3 used station-level onsets. For the prediction tasks (RQ4–RQ5), the definition was applied to each station separately: runs separated by at most 3 observed hours at or below 75 $\mu g/m^3$ were merged, and gaps containing missing data were not bridged. Episodes whose start or end lay within the merge window of missing data were labelled "uncertain" and excluded from labelling and evaluation. Episode counts therefore differ between the two rules and are not directly comparable.

3.3 Analytical methods

3.3.1 RQ1

PM2.5 was summarised by mean, median, and 90th percentile (P90), by month and by hour of day within season. Regional episodes were described by duration, peak tier (below 150, 150–250, and at least 250 $\mu g/m^3$), start month, and start hour. For each station, the difference between its daily mean PM2.5 and the 12-station daily mean was estimated, with 95% confidence intervals (CIs) from a block bootstrap (14-day blocks, 2,000 resamples).

3.3.2 RQ2

Spearman's rank correlation (ρ) between PM2.5 and each co-pollutant and weather variable was computed for the pooled data and for each station. Lagged correlations between each variable at time $t - k$ and PM2.5 at time t (k = 0–24 h) were computed within stations. High hours (PM2.5 $> 75\ \mu g/m^3$) were compared with all other hours using the rank-biserial correlation ($r = 2 \cdot \mathrm{AUC} - 1$; $|r| < 0.1$ negligible, < 0.3 small, < 0.5 medium, otherwise large) and a two-sided Mann–Whitney U test, without correction for multiple comparisons. The results are associations, not causal effects.

3.3.3 RQ3

The five weather variables were standardised and clustered with K-means (K = 2–8) on hours with complete weather data. K was the value with the highest silhouette score (5,000-hour random subsample), provided the smallest cluster held at least 5% of hours. Stability was assessed by the adjusted Rand index across five random seeds. Regimes were ranked by mean PM2.5 (R1 highest), and their association with season was measured by Cramér's V. The onset ratio of a regime was its share of station-level onsets divided by its share of hours (1 = proportional), with 95% CIs from a 5-day block bootstrap (2,000 resamples). A regime was considered to separate onsets if its CI excluded 1 and $|\text{ratio} - 1| \geq 0.10$; regimes with fewer than 30 onsets were not evaluated. One of the 6,002 station-level onsets had no weather data at the onset hour and was excluded, leaving 6,001.

3.3.4 RQ4

To address RQ4, five prediction horizons ($H = 1, 3, 6, 12$, and 24 hours) were tested, with $H = 6$ selected as the primary horizon. Predictions were generated only when PM2.5 ≤ 75 μg/m^3 and no episode was active. The target was labelled 1 if a new episode began within the next H hours. Input features included calendar, meteorological, co-pollutant, and PM2.5 variables (current values, lags, and changes), using only information available up to time t and excluding station identifiers. Seven input scenarios were compared using LightGBM, with early stopping based on validation PR-AUC. Data were split chronologically into training (March 2013–February 2015), validation (March 2015–February 2016), and test (March 2016–February 2017), with 7-day gaps between sets to reduce data leakage. Model configurations and alert thresholds were selected on validation, while the test set was reserved for final evaluation. Performance was assessed using episode recall, usable recall (the proportion of episodes detected at least 3 hours before onset), false alerts per station per 30 days, median lead time, and PR-AUC lift over climatology. A warning was considered a false alarm if no valid episode onset occurred within the next H hours. Once a warning was issued, the station remained silent for H hours. 95% confidence intervals were estimated using 2,000 paired 5-day block bootstrap iterations.
The primary comparison was between model B and the PM2.5-only model at $H = 6$. An improvement was considered statistically supported when the lower bound of the 95% confidence interval for the difference exceeded zero. Other comparisons were exploratory, with model C compared against B to assess potential data leakage. For deployment models at $H = 6$ and 12, alert thresholds were selected on validation to maximise usable recall while limiting false alarms to 12 per station per 30 days. When the difference in usable recall between models was below 0.02, the simpler model was preferred. The selected models were then evaluated on the test set against three criteria: (G1) usable recall above climatology, with a lower 95% CI bound greater than zero; (G2) no more than 15 false alarms per station per 30 days; and (G3) a median lead time of at least 3 hours. Finally, feature importance was assessed using permutation importance separately for each station.

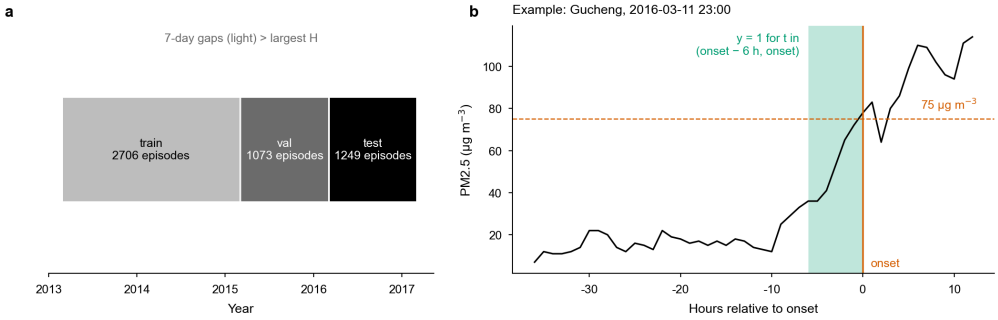


Figure 2: Chronological data split and onset labelling

3.3.5 RQ5

For RQ5, the model's generalizability to previously unseen monitoring stations was assessed using Leave-One-Station-Out (LOSO) validation. Specifically, each model was trained on 11 stations and tested on the remaining held-out station, repeating the process across all 12 stations. Early stopping and prediction thresholds were selected using the validation data from the 11 training stations. LOSO results were compared with the pooled model through paired differences in usable recall and false alarms. A station was considered to pass the transfer criterion if its LOSO false alarms were within 1.25 times the allowed budget and its recall exceeded the station-specific climatology. Additionally, `station_id` was included as a predictor, and LOSO was repeated at $H = 24$ hours as supplementary checks.

3.4 Cost model

Missed episodes (false negatives) and false alarms (false positives) are the two error types of an early-warning system. As a secondary metric, we assumed a hypothetical cost ratio of 10:1 between a missed episode and a false alarm, following cost-sensitive learning [2]:

$$\text{Total Cost} = 10 \times \text{Missed Episodes} + \text{False Alarms}. \tag{1}$$

A missed episode was one without a warning ≥ 1 h before onset, and costs were expressed per station per 30 days. For this metric only, the alert threshold was the value that minimised the total cost on the validation set, without the 12-false-alarm cap used for the primary comparisons. The ratio is an assumption, not an empirical economic cost.

4 Results

The 95% confidence intervals (CIs) are from a block bootstrap. The comparison of B with PM2.5-only (Section 4.2) and the deployment selection rule (Section 4.4) were pre-specified; all other comparisons are exploratory. Model scenarios are defined in Table 1.

4.1 PM2.5 peaked in winter, but episodes occurred year-round and differed between stations

A total of 463 regional PM2.5 episodes were detected, with a median duration of 19 h (P90 = 66 h, longest 139 h; Table 2). Monthly mean PM2.5 was highest in December (105 $\mu g/m^3$) and lowest in August (53 $\mu g/m^3$; Fig. 3a). By hour of day, mean PM2.5 was highest at 22:00 and lowest at 07:00 (Fig. 3b). Of all episodes, 49% began between October and March, and episodes began in all 12 months (Fig. 3d). Seasonality was clearer among episodes with a peak ≥ 250 $\mu g/m^3$: 85% of these began between October and March (Table 2). Station means ranged from 66.0 $\mu g/m^3$ (Dingling) to 86.2 $\mu g/m^3$ (Dongsi); the CI of the difference from the 12-station mean excluded 0 at every station except Shunyi (Fig. 3c).

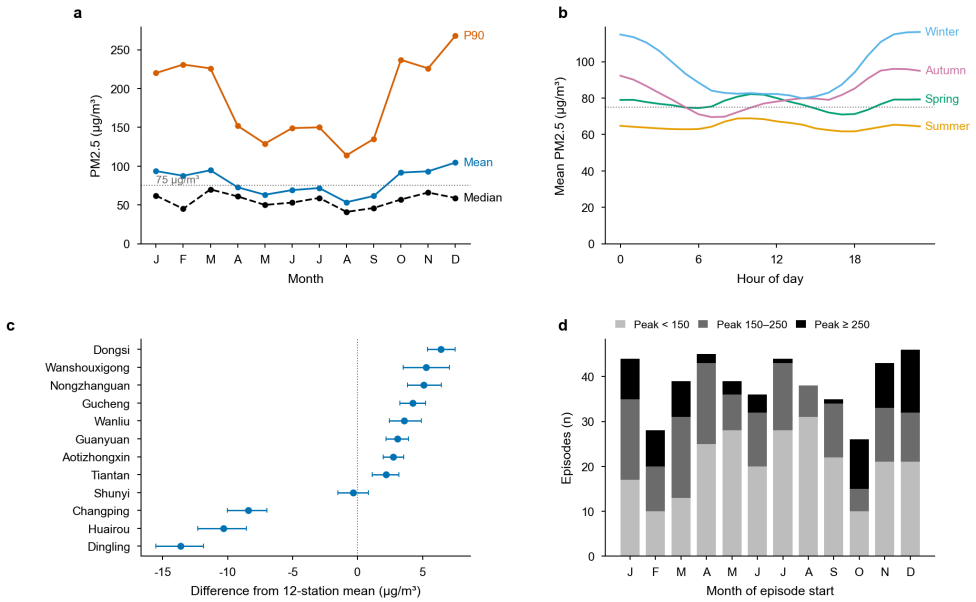


Figure 3: Seasonal, diurnal and station patterns of PM2.5 and episode onsets

Table 2: Regional PM2.5 episodes by peak concentration

Group	n	Share (%)	Duration, median [P90] (h)	Longest (h)	Peak, median (μg/m^3)	Started Oct–Mar (%)	Started 18–22 h (%)
All episodes	463	100.0	19 [66]	139	144	49	36
Peak < 150	246	53.1	11 [25]	78	110	37	37
Peak 150–250	146	31.5	30 [57]	107	184	51	33
Peak ≥ 250	71	15.3	67 [113]	139	322	85	38

Note. Episode: regional mean PM2.5 $>$ 75 μg/m^3 for ≥ 6 h; missing gaps ≤ 3 h bridged. $n = 463$ episodes. P90, 90th percentile.

Across 412,029 station-hours with PM2.5 observed (of 420,768; 2.1% missing), PM2.5 had strong Spearman rank correlations with PM10 ($\rho = +0.89$), CO (+0.84), NO2 (+0.66), and SO2 (+0.50), with the same sign at all 12 stations (Fig. 4a, Table 3). PM10 contains PM2.5 by the definition of the measurement. All weather variables had $|\rho| \leq 0.33$ (wind speed −0.33, dew point +0.24; remaining variables $|\rho| \leq 0.07$). Comparing hours with PM2.5 > 75 $\mu g/m^3$ ($n = 161,002$) with all other hours ($n = 251,027$), the rank-biserial coefficient was +0.89 (PM10), +0.83 (CO), +0.62 (NO2), +0.50 (SO2), −0.24 (O3), −0.28 (wind speed), and +0.18 (dew point), while pressure, temperature, and rain had $|r| < 0.1$; $P < 0.001$ for all variables (Fig. 4b, Table 3). At a lag of 24 h, the autocorrelation of PM2.5 was $\rho = +0.36$ (Fig. 4c).

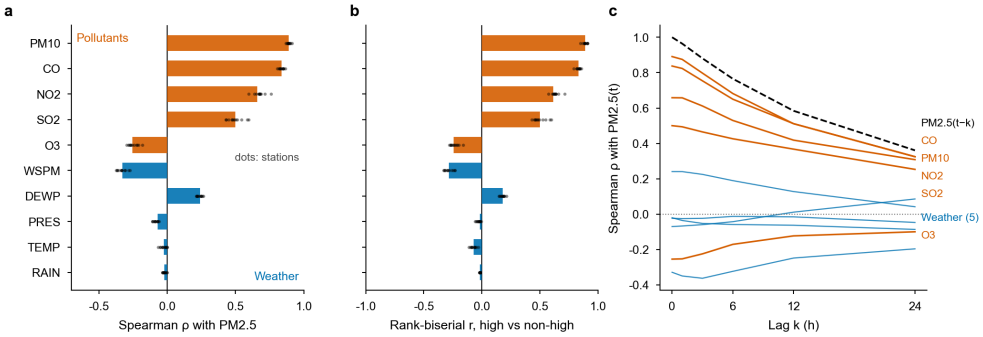


Figure 4: Association of PM2.5 with co-pollutants and weather

Table 3: Association of PM2.5 with co-pollutants and weather

Variable	Group	Spearman ρ	Station range	Same sign (stations)	Rank-biserial r	Effect size	P
PM10 (μg/m^3)	Pollutant	+0.89	+0.87 to +0.91	12/12	+0.89	large	< 0.001
CO (μg/m^3)	Pollutant	+0.84	+0.80 to +0.86	12/12	+0.83	large	< 0.001
NO2 (μg/m^3)	Pollutant	+0.66	+0.60 to +0.76	12/12	+0.62	large	< 0.001
SO2 (μg/m^3)	Pollutant	+0.50	+0.43 to +0.60	12/12	+0.50	large	< 0.001
O3 (μg/m^3)	Pollutant	-0.25	-0.30 to -0.18	12/12	-0.24	small	< 0.001
WSPM (m/s)	Weather	-0.33	-0.37 to -0.27	12/12	-0.28	small	< 0.001
DEWP (°C)	Weather	+0.24	+0.21 to +0.26	12/12	+0.18	small	< 0.001
PRES (hPa)	Weather	-0.07	-0.11 to -0.06	12/12	-0.02	negligible	< 0.001
TEMP (°C)	Weather	-0.02	-0.07 to +0.00	10/12	-0.07	negligible	< 0.001
RAIN (mm)	Weather	-0.02	-0.03 to -0.00	12/12	-0.02	negligible	< 0.001

Note. Spearman ρ with hourly PM2.5, all stations pooled ($n = 412,029$); station range, min to max over 12 stations. Rank-biserial r compares high (PM2.5 > 75; $n = 161,002$) with non-high hours ($n = 251,027$); two-sided Mann–Whitney U, no multiplicity correction. PM10 contains PM2.5 by construction. Associations, not causal effects.

K-means on the five weather variables selected $K = 2$ (silhouette 0.425; Fig. 5a). $K = 3$ had a higher silhouette (0.430), but its smallest cluster contained only 0.2% of hours, below the 5% threshold, so it was excluded. The solution was stable across seeds (ARI 0.996–0.998). Regime R1 (cold, high-pressure, dry) contained 47.5% of hours with mean PM2.5 of 83.8 $\mu g/m^3$; R2 (warm, low-pressure, humid) contained 52.5% with 76.2 $\mu g/m^3$ (Table 4). The two regimes were closely tied to season (Cramér's V = 0.71; Fig. 5d). The onset ratio of a regime is the share of onsets falling in the regime divided by the regime's share of hours; a regime was considered to separate onsets if the CI of the ratio excluded 1 and $|\text{ratio} - 1| \geq 0.10$. The onset ratio was 0.93 [0.86, 1.00] for R1 (2,653 onsets) and 1.06 [1.00, 1.13] for R2 (3,348 onsets), so neither regime met the criterion (Fig. 5c).

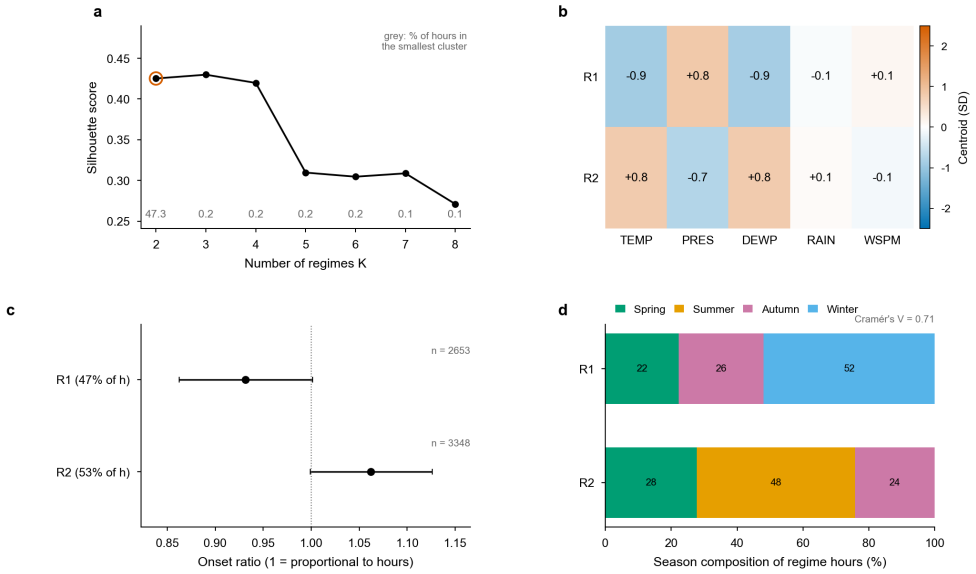


Figure 5: Weather regimes and episode onsets

Table 4: Weather Regimes: PM2.5 level and episode onsets

Regime	Weather (centroid > 0.5 SD)	Hours (%)	PM2.5 mean [P90] (μg/m^3)	High hours (%)	Onsets (n)	Onset ratio [95% CI]
R1	cold, high-pressure, dry	47.5	83.8 [214]	38.6	2653	0.93 [0.86–1.00]
R2	warm, low-pressure, humid	52.5	76.2 [162]	39.5	3348	1.06 [1.00–1.13]

Note. K-means on standardised TEMP, PRES, DEWP, RAIN, WSPM ($K = 2$ by silhouette; $n = 420,359$ h). Regimes ordered by mean PM2.5. Onset ratio = share of the 6,001 station-level onsets ÷ share of hours in the regime (1 = proportional); 95% CI, 5-day block bootstrap, 2000 resamples; n/a if a regime has < 30 onsets. High hour, PM2.5 > 75 μg/m^3.

4.2 Current PM2.5 was the main source of information for anticipating onset

The pre-specified criterion was that scenario B (co-pollutants + weather, without PM2.5) had to exceed PM2.5-only (PM2.5 at hours t and $t-1$) in both PR-AUC lift and recall at the same false-alarm budget. This criterion was not met. At $H = 6$ h (59,545 test samples, 12.1% positive, 1,249 episodes), the difference B − PM2.5-only was −0.050 [−0.322, 0.197] in PR-AUC lift and −0.196 [−0.253, −0.146] in recall (Fig. 6c); recall here is the share of episodes with at least one warning ≥ 1 h before onset. Each model's threshold was chosen on validation to allow at most 12 false alarms per station per 30 days; on the test year, false alarms ranged from 9.9 to 12.4 (Table 5).

Recall was 0.90 [0.87, 0.93] for PM2.5-only and PW (W + PM2.5), but only 0.70 [0.65, 0.76] for B (Table 5, Fig. 6b). In exploratory comparisons, W exceeded climatology by +0.290 [0.222, 0.360] in recall and +0.920 [0.612, 1.291] in PR-AUC lift; B exceeded A by +0.208 [0.154, 0.264] in recall; and adding PM2.5 to B (C − B) raised recall by +0.167 [0.133, 0.212]. PW did not differ from PM2.5-only in recall (+0.002 [−0.031, 0.031]) but had a higher PR-AUC lift (+0.410 [0.188, 0.648]).

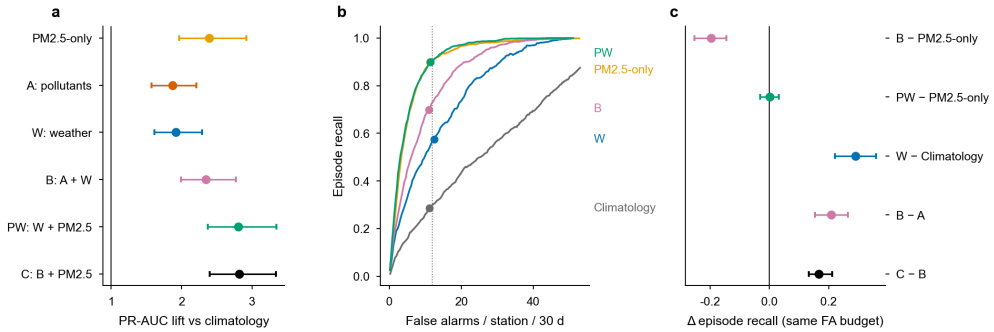


Figure 6: Scenario comparison at H = 6 h

Table 5: Test performance at H = 6 h (Value [95% Block-bootstrap CI])

Model	PR-AUC lift vs climatology	Episode recall	False alarms / station / 30 d	Median lead (h)	Cost (10:1) / station / 30 d
Climatology	1.00 [1.00, 1.00]	0.29 [0.23, 0.34]	11.0 [9.4, 12.7]	4	68.7 [63.0, 74.5]
PM2.5-only	2.39 [1.96, 2.92]	0.90 [0.87, 0.93]	11.4 [9.5, 13.5]	2	20.6 [17.4, 23.8]
A: pollutants	1.87 [1.57, 2.21]	0.49 [0.43, 0.56]	9.9 [7.9, 12.0]	3	37.4 [33.6, 41.3]
W: weather	1.92 [1.61, 2.29]	0.57 [0.51, 0.64]	12.4 [10.4, 14.8]	3	39.6 [34.7, 44.6]
B: A + W	2.34 [1.99, 2.77]	0.70 [0.65, 0.76]	11.2 [9.2, 13.2]	4	31.1 [26.7, 35.7]
PW: W + PM2.5	2.80 [2.37, 3.35]	0.90 [0.87, 0.93]	11.5 [9.7, 13.4]	3	20.1 [17.1, 23.4]
C: B + PM2.5	2.82 [2.40, 3.34]	0.87 [0.83, 0.90]	10.9 [9.1, 12.9]	3	21.6 [18.2, 25.3]

Note. $n = 1249$ test episodes, 12 stations, 59545 samples. Recall and false alarms use the validation threshold giving at most 12 false alarms/station/30 d; cost uses the 10:1 threshold. Climatology lift = 1 by definition.

4.3 The advantage of additional information shrank as the forecast horizon grew

As H increased from 1 h to 24 h, the positive rate in the test set rose from 0.020 to 0.369. Recall (warning $\geq$ 1 h ahead) of PM2.5-only rose from 0.55 [0.51, 0.59] to 0.96 [0.95, 0.97], PR-AUC lift over climatology fell from 9.43 [7.64, 11.26] to 1.50 [1.33, 1.67], and the median lead time rose from 1 h to 11 h (Fig. 7). The lift of W fell from 2.53 [2.05, 3.22] at $H = 1$ h to 1.50 [1.34, 1.68] at $H = 24$ h, equal to the lift of PM2.5-only at $H = 24$ h. The recall difference PW − PM2.5-only was +0.021 [−0.016, 0.056] at $H = 1$ h, −0.017 [−0.053, 0.019] at 3 h, +0.002 [−0.028, 0.033] at 6 h, +0.012 [0.000, 0.027] at 12 h, and +0.003 [−0.004, 0.011] at 24 h; the CI included 0 at $H = 1$, 3, 6, and 24 h (Fig. 7b).

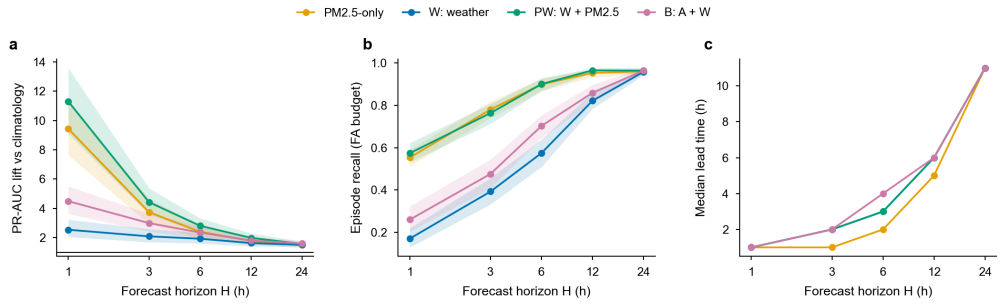


Figure 7: Effect of forecast horizon

4.4 Deployment model: PM2.5 + weather, 12 h ahead

Candidates were the scenarios at $H = 6$ h and $H = 12$ h. On validation (thresholds allowing at most 12 false alarms per station per 30 days), usable recall, i.e. the share of episodes warned $\geq$ 3 h before onset, was highest for C at $H = 12$ h (0.779). PW at $H = 12$ h reached 0.775, within 0.02 of the best;

PM2.5-only reached 0.757, below the 0.759 cut-off, and was not eligible. The pre-specified rule selects the simplest eligible model, which was PW at $H = 12$ h (Table 6, Fig. 8a). At $H = 12$ h, usable recall was higher than at $H = 6$ h for every model.

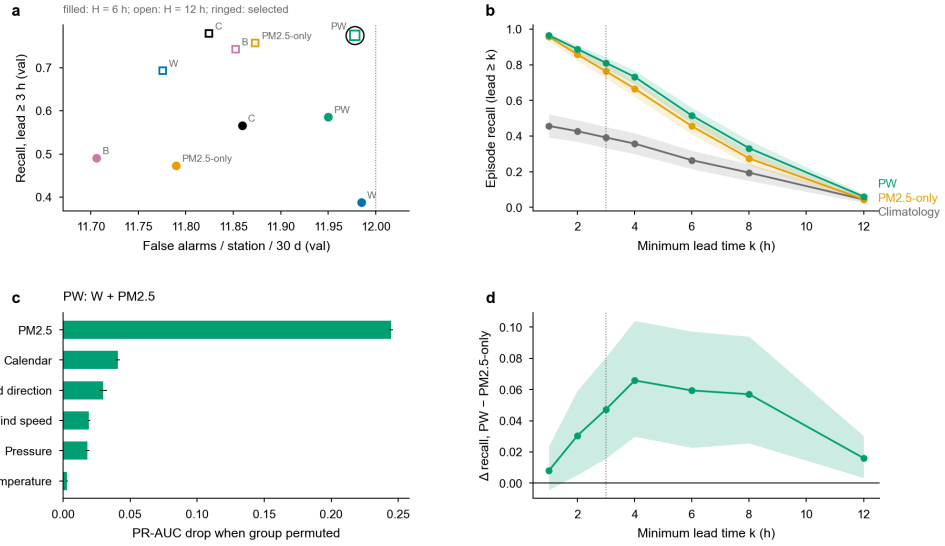


Figure 8: Selection and test performance of the deployment model

Table 6: Deployment candidates on validation (Selection data only)

Selected	H (h)	Model	Recall, lead ≥ 3 h	Recall, any lead	False alarms / station / 30 d	Median lead (h)	PR-AUC
	6	Climatology	0.226	0.327	11.957	4.0	0.159
	6	PM2.5-only	0.473	0.914	11.79	3.0	0.383
	6	W: weather	0.388	0.61	11.985	3.0	0.312
	6	PW: W + PM2.5	0.585	0.906	11.95	3.0	0.449
	6	B: A + W	0.49	0.727	11.706	4.0	0.383
	6	C: B + PM2.5	0.566	0.897	11.859	3.0	0.455
	12	Climatology	0.442	0.503	11.929	7.0	0.238
	12	PM2.5-only	0.757	0.961	11.873	5.0	0.444
	12	W: weather	0.693	0.806	11.776	6.0	0.409
◀	12	PW: W + PM2.5	0.775	0.958	11.978	6.0	0.497
	12	B: A + W	0.743	0.868	11.852	6.0	0.481
	12	C: B + PM2.5	0.779	0.954	11.825	6.0	0.514

Note. Threshold per candidate: most episodes detected with ≤ 12 false alarms/station/30 d. Rule: best useful recall, then the simplest model within 0.02. The arrow marks the selected deployment candidate.

On 1,246 episodes, PW achieved a usable recall of 0.81 [0.77, 0.84], 11.3 [9.7, 13.0] false alarms per station per 30 days, a median lead time of 6 h, and a PR-AUC of 0.561 [0.506, 0.622]. PM2.5-only reached 0.76 [0.72, 0.80] with 11.2 [9.3, 12.9] false alarms, and climatology 0.39 [0.33, 0.45] with 10.8 [9.3, 12.5]. PW exceeded climatology by +0.419 [0.349, 0.492] in usable recall (Table 7). Compared with PM2.5-only, PW was higher by +0.047 [0.015, 0.081] in usable recall (exploratory comparison). With no minimum lead, PW reached a recall of 0.97 [0.95, 0.98], and the difference from PM2.5-only was +0.008 [−0.004, 0.023]; false alarms did not differ (+0.2 [−0.6, 0.9]; Table 7). PW passed all three deployment gates: the lower CI bound of the recall difference from climatology exceeded 0, false alarms (11.3) did not exceed the limit of 15, and the median lead time (6 h) exceeded the 3 h minimum.

Table 7: Paired differences for the selected model (Test, H = 12 h)

Comparison	Metric	Difference [95% CI]
PW: W + PM2.5 − PM2.5-only	Recall, lead ≥ 3 h	0.047 [0.015, 0.081]
PW: W + PM2.5 − PM2.5-only	Recall, any lead	0.008 [−0.004, 0.023]
PW: W + PM2.5 − PM2.5-only	False alarms / station / 30 d	0.2 [−0.6, 0.9]
PW: W + PM2.5 − Climatology	Recall, lead ≥ 3 h	0.419 [0.349, 0.492]
PW: W + PM2.5 − Climatology	Recall, any lead	0.509 [0.442, 0.575]

Note. Differences computed within the same block-bootstrap resamples. A positive recall difference favours the selected model; lower false alarms are better.

The PM2.5 feature group was the most important (PR-AUC fell by 0.244 when permuted), followed by calendar (0.041), wind direction (0.030), and the remaining groups (≤ 0.019; Fig. 8c). The recall difference PW − PM2.5-only by minimum lead k ranged from +0.030 to +0.066 at k = 2–8 h (all CIs excluded 0) and was +0.016 [0.003, 0.030] at 12 h; it was +0.008 [−0.004, 0.024] at k = 1 h, the only lead at which the CI included 0 (Fig. 8d). Recall of both PW and PM2.5-only decreased as k increased (Fig. 8b).
In the test year, 43 of 1,246 episodes (3.5%) received no warning, and there were 1,622 false alarms (11.3 per station per 30 days). Fig. 9 shows three cases selected by the rules in its caption, not by appearance: a typical hit with a 7 h lead (lead time closest to the median; Aotizhongxin, 15 April 2016; panel a), the most severe miss (Gucheng, 15 December 2016; panel b) and the highest-scoring false alarm (Shunyi, 21 January 2017; panel c).

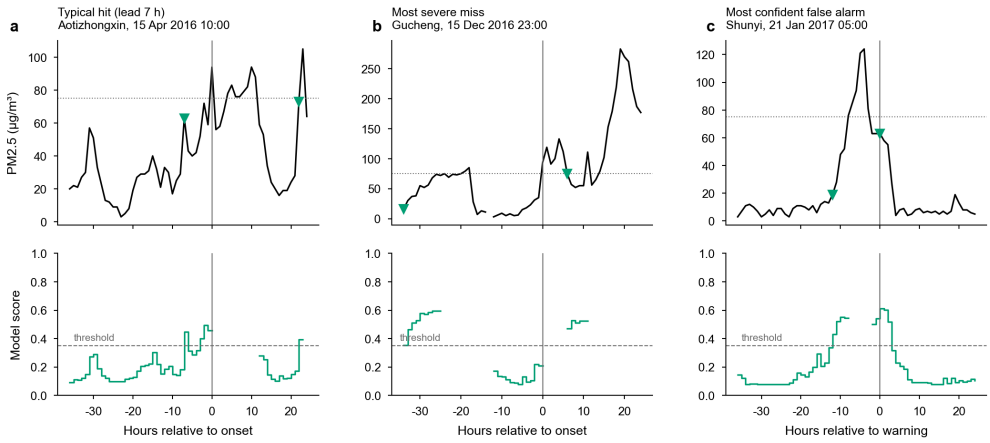


Figure 9: Typical hit, most severe miss and most confident false alarm

4.5 The model transferred to unseen stations with a small loss

At H = 12 h, the usable recall of PW on unseen stations (leave-one-station-out, LOSO) was lower than that of the pooled model by −0.013 [−0.025, −0.001] (0.80 vs 0.81), and lower at 8 of 12 stations. False alarms were 11.1 vs 11.3 (Δ = −0.20 [−0.39, −0.01]). For PM2.5-only (+0.002 [−0.004, 0.007]), W (−0.010 [−0.030, 0.008]), and climatology (−0.026 [−0.077, 0.021]), the CI included 0 (Table 8, Fig. 10c). Compared with LOSO climatology, PW was higher by +0.432 [0.342, 0.528].

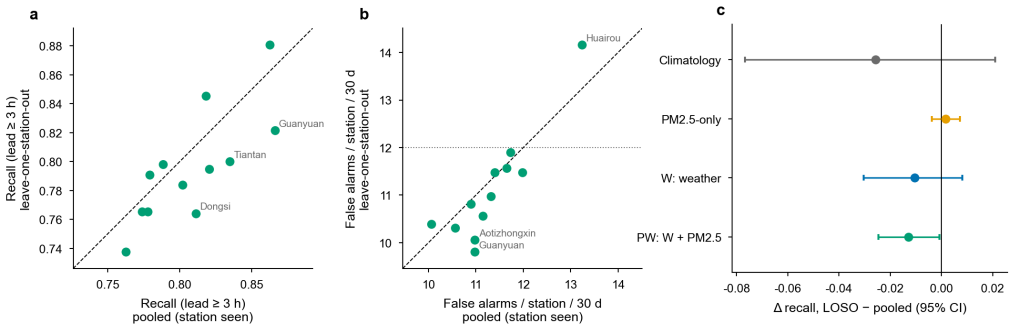


Figure 10: Transfer to unseen stations (leave-one-station-out)

Table 8: Leave-one-station-out vs pooled model (Test, H = 12 h, Recall with lead ≥ 3 h)

Model	Pooled recall	LOSO recall	Δ recall [95% CI]	Pooled FA	LOSO FA	Δ FA [95% CI]	Stations LOSO recall < pooled
Climatology	0.39	0.37	-0.026 [-0.077, 0.021]	10.8	10.6	-0.3 [-1.6, 1.0]	7/12
PM2.5-only	0.76	0.76	0.002 [-0.004, 0.007]	11.2	11.2	0.0 [-0.0, 0.1]	2/12
W: weather	0.71	0.70	-0.010 [-0.030, 0.008]	11.9	12.1	0.2 [-0.1, 0.5]	8/12
PW: W + PM2.5	0.81	0.80	-0.013 [-0.025, -0.001]	11.3	11.1	-0.2 [-0.4, -0.0]	8/12

Note. FA = false alarms/station/30 d. Δ = LOSO − pooled, 5-day block bootstrap shared across stations. Thresholds for LOSO were set on the 11 training stations only.

All 12 stations passed the transfer gate: LOSO false alarms did not exceed 15 and recall exceeded the LOSO climatology of the same station (Table 9). LOSO false alarms ranged from 9.8 to 14.2 per station per 30 days; Huairou (14.2) was the only station above the budget of 12 but still below the limit of 15. LOSO recall of PW ranged from 0.74 to 0.88 across stations, compared with 0.21–0.48 for LOSO climatology (Fig. 10a–b, Table 9).

Table 9: PW: W + PM2.5 on unseen stations (LOSO, H = 12 h)

Station	Episodes	Pooled recall	LOSO recall	Climatology recall	LOSO FA	Transfer gate
Aotizhongxin	110	0.82	0.85	0.35	10.1	pass
Changping	81	0.78	0.77	0.25	11.0	pass
Dingling	80	0.76	0.74	0.21	11.6	pass
Dongsi	106	0.81	0.76	0.41	10.3	pass
Guanyuan	112	0.87	0.82	0.38	9.8	pass
Gucheng	104	0.79	0.8	0.45	10.8	pass
Huairou	86	0.78	0.79	0.38	14.2	pass
Nongzhanguan	111	0.8	0.78	0.31	10.6	pass
Shunyi	117	0.82	0.79	0.48	10.4	pass
Tiantan	115	0.83	0.8	0.33	11.5	pass
Wanliu	109	0.86	0.88	0.38	11.5	pass
Wanshouxigong	115	0.77	0.77	0.4	11.9	pass

Note. Gate: LOSO false alarms $\leq 1.25 \times 12$ and recall above LOSO climatology at that station.

Adding `station_id` to PW made no difference in recall (−0.016 [−0.045, 0.015]), false alarms (−0.2 [−0.8, 0.5]), or PR-AUC (+0.021 [−0.003, 0.048]). At $H = 24$ h, LOSO and pooled lifts were similar: PW 1.54 [1.39, 1.70] vs 1.56 [1.40, 1.74] ($\Delta = -0.02$ [−0.06, 0.01]), W 1.51 vs 1.50, and PM2.5-only 1.49 vs 1.50. The lift of LOSO climatology was lower than that of the pooled model (0.89 [0.85, 0.94] vs 1.00; $\Delta = -0.11$ [−0.15, −0.06]).

5 Discussion

5.1 Summary of main findings

This study evaluated how well the onset of PM2.5 episodes can be anticipated from observations at 12 Beijing stations. The central finding is that current PM2.5 clearly dominates: the pre-specified hypothesis that co-pollutants and weather without PM2.5 (B) would outperform the PM2.5-only model was not supported. Weather nonetheless added a small but consistent amount of information at intermediate lead times. The deployment model selected by the pre-specified rule reached a usable recall of 0.811 on the test year, passed all three deployment gates, and transferred to unseen stations with a small loss. Together, these findings indicate that recent PM2.5 measurements carry most of the signal for episode onset, while weather variables add limited value.

5.2 Why current PM2.5 dominates

The results indicate that recent PM2.5 measurements provide the strongest predictive signal for episode onset. The PM2.5-only model achieved a recall of 0.90 at H = 6 h, compared with 0.70 for Model B, suggesting that co-pollutants and weather alone cannot capture the immediate changes preceding an episode as effectively as PM2.5 itself. The strong correlations between PM2.5 and co-pollutants, particularly CO and NO2, further suggest that these variables capture overlapping information associated with common emission sources and dispersion conditions rather than provide entirely independent predictive signals.

5.3 The indirect role of weather

Weather provides meaningful predictive information, as Model W outperformed climatology by +0.290 in recall and +0.920 in PR-AUC lift. However, the weak univariate correlations ($|\rho| \leq 0.33$), which are in line with [16], who showed that the effect of wind is not simply linear, and the failure of the two meteorological regimes to distinguish episode onsets, despite their strong association with season (Cramér's V = 0.71), suggest that broad weather patterns influence baseline PM2.5 levels more than the timing of onset. Instead, onset may depend on short-term processes, such as pollutant transport, boundary-layer variability, and wind-direction shifts, which station-level measurements capture only partially. Although these mechanisms were not directly tested, this interpretation is consistent with the permutation analysis, in which wind direction and calendar were the most informative feature groups after PM2.5 (Section 4.4).

5.4 Effect of forecast horizon

As the prediction horizon increased from 1 to 24 hours, the positive rate rose from 0.020 to 0.369, while the advantage of additional predictors diminished. The lift of Model W declined from 2.53 to 1.50, equal to that of the PM2.5-only model at H = 24 h, and the recall difference between PW and PM2.5-only included zero at H = 1, 3, 6, and 24 h. This suggests that longer-horizon predictions increasingly reflect seasonal patterns and baseline pollution levels rather than conditions immediately preceding onset. The higher positive rate also makes conventional recall less discriminative, because frequent warnings are more likely to coincide with an onset without giving sufficient advance notice. Since precision–recall-based measures are preferred over single recall values for imbalanced data [11], usable recall (warning at least 3 hours ahead) and lift are more informative criteria for model selection than conventional recall alone.

5.5 Implications and limits of the deployment model

The improvement of PW over the PM2.5-only model should be interpreted cautiously. PW achieved only a modest increase in usable recall (+0.047, 95% CI [0.015, 0.081]), and this comparison was exploratory. The advantage was clearest when a minimum lead time of 2–8 hours was required (+0.030 to +0.066) but disappeared when warnings with only 1 hour of lead time were accepted, suggesting that PM2.5 alone detects most onsets if late warnings are considered sufficient. Selecting PW also reflects the predefined rule of choosing the simplest eligible model, not evidence that PM2.5-only performs substantially worse: PM2.5-only missed the validation cut-off by just 0.002 (0.757 vs. 0.759), and the best validation model, C, exceeded PW by only 0.004. H = 12 h is a reasonable trade-off, as it gave higher usable recall than H = 6 h for every model and a median lead time of 6 hours for response preparation. The false-alarm

budget and the 10:1 cost ratio are assumptions rather than empirical costs, following a cost-sensitive approach [2].

5.6 Error analysis

Errors fall into three groups; case selection rules are given in the caption. First, 43 of 1,246 episodes (3.5%) received no warning. The most severe miss, at Gucheng on 15 December 2016, was an abrupt onset after which PM2.5 reached 302 μg/m^3 within 24 h, which suggests long-range transport that local observations had not yet reflected. Second, about 16% of episodes were warned late (lead time below 3 h), as shown by the gap between any-lead recall and usable recall. Third, the 1,622 false alarms (11.3 per station per 30 days) include cases such as Shunyi on 21 January 2017, where PM2.5 briefly exceeded 75 μg/m^3 but fell before the 6-hour duration required to count as an episode. Some false alarms therefore reflect the strict episode definition rather than complete model failure. Reducing both abrupt misses and transient false alarms may require macro-scale data such as boundary-layer height, upstream observations or numerical weather forecasts. This analysis is based on illustrative cases rather than a full census of failure modes.

5.7 Transferability to new stations

In leave-one-station-out (LOSO) evaluation, the usable recall of PW decreased by only -0.013 [-0.025, -0.001] relative to the pooled model (0.80 vs. 0.81), all 12 stations passed the transfer gate, and the recall of PW exceeded LOSO climatology by $+0.432$. Adding `station_id` did not improve recall, false alarms or PR-AUC, indicating that the model does not need to learn station identity and could in principle be applied to a new station in the same city without retraining. An important caveat is that all 12 stations lie in the same city, share the same climate and emission sources, and observe the same pollution events at the same time. Because spatial dependence can make validation that mixes locations overly optimistic [6, 9], LOSO is a stricter test than random splitting; nevertheless, it tests transfer only between stations within Beijing and is likely to remain more optimistic than transfer to a city with different emission characteristics or climate.

5.8 Limitations

This study has several limitations. First, the data come from a single city with a single test year, so generalization to other years or regions has not been confirmed. Second, the results depend on the definition of onset, the 75 μg/m^3 threshold and the false-alarm budget; changing these choices could alter model rankings. Third, inputs were limited to station observations, without boundary-layer height, upstream data or meteorological forecasts. Fourth, only two comparisons were pre-specified; the remainder are exploratory and not corrected for multiple comparisons, and should be treated as hypothesis-generating rather than confirmatory. Fifth, the associations are predictive, not causal, and the mechanisms discussed above were not tested directly. Finally, emission structure changes over time, so the model should be re-validated with new data before deployment.

6 Conclusion

This study examined whether the onset of PM2.5 episodes in Beijing can be anticipated from station observations. Current PM2.5 and its recent change carried most of the predictive information: a PM2.5-only model reached an episode recall of 0.90 at $H = 6$ h, against 0.70 for co-pollutants and weather without PM2.5, so the pre-specified hypothesis was not supported. Weather added modest, horizon-dependent value. The selected PM2.5-plus-weather model at $H = 12$ h reached a usable recall of 0.81 with 11.3 false alarms per station per 30 days and passed the three pre-specified gates, but exceeded PM2.5-only by only 0.047 in an exploratory comparison. Weather regimes were closely tied to season yet did not separate onsets. On held-out stations, usable recall fell by only 0.013, suggesting that retraining may not be needed for new stations in the same city, although Huairou slightly exceeded the false-alarm budget. These results are limited by a single city, a single test year, dependence on the 75 μg/m^3 threshold and the false-alarm budget, and mostly exploratory comparisons. Future work should test other years and cities and add boundary-layer height, upstream observations and weather forecasts, which may help reduce abrupt misses and transient false alarms.

AUTHORS CONTRIBUTIONS

Member	Contribution
Nguyen Nguyet Anh 11230733	• Identified and shaped the research questions, defining the master question and five linked questions, from describing PM2.5 patterns to predicting episode onset and testing transfer to unseen stations. • Developed the analysis plan for RQ4 and RQ5, including the pre-specified hypothesis, input scenarios, chronological split with gaps to prevent leakage, evaluation metrics, false-alarm budget and leave-one-station-out design. • Analysed and interpreted the RQ4 results, comparing scenarios and forecast horizons, separating the pre-specified hypothesis from exploratory findings, and examining the deployment model. • Writing the Results section, reporting the findings for all five research questions with confidence intervals and cross-checking the numbers against the analysis outputs.
Vu Phuong Anh 11230736	• Cleaned and preprocessed the dataset, conducted EDA, and examined PM2.5 patterns across 12 monitoring stations. • Wrote the Methods section to describe data preprocessing procedures, analytical approaches, modelling techniques, and validation methods. • Addressed RQ1, RQ4, and RQ5 through relevant analyses and interpretation of the research findings. • Prepared the presentation slides.
Vu Khanh Linh 11230760	• Conducted exploratory data analysis (EDA) and data visualization. • Developed research questions and hypotheses. • Designed charts for the presentation. • Participated in numerical analysis to explain the research questions. • Summarized and compared analytical results.
Tran Thu Thuy 11230780	• Analyzed the data and provided feedback on the business problem and research questions. • Co-developed the content for RQ2 and RQ3, including the analysis and interpretation of results. • Wrote the Abstract and Conclusion sections. • Reviewed and revised the full manuscript to ensure consistency and accuracy. • Prepared the presentation slides.

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